# LArPix ASICs on PACMAN

Bench setup, configuration, replies and status

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## 1 Purpose and scope

This note records the LArPix ASIC work on PACMAN with the zipline firmware and software (`zipline.tex`): which ASICs are on the bench, how they are configured, what they reply, and what is open. Register numbers are those of the LArPix v3 register map in `larpix-lib` (`config/asics/larpix_v3.yaml`), which agrees with the LBNL datasheet (rev 2.4, not public). The datasheet has known errors: its bit layouts and defaults are not trusted until they are read back from a chip. The control software, `larpix-lib`, has its own record (larpix-lib `agents/larpix-lib.tex`).

## 2 Bench setup

- One LArPix v3 on a single-tile board, on tile 1. On the 1v4 it is reached through the J1302 cable, with all four PISO/POSI pairs routed to the tile-1 UARTs 0–3.
- Tile-1 UART 0 drives the ASIC's POSI1 (its west input) and receives PISO0 (west output): with only POSI1 enabled, requests on UARTs 1–3 go unanswered (2026-11-02).
- Chip ID 0x0B (11). The hello request is a configuration read of register 122 (the chip ID); the expected reply is `0x42254139 1C2DE82F` ("read, chip 0x0B, register 122, value 0x0B").
- Tile power by `pacman_menu` (VDDA about 2.32 V, VDDD about 1.20–1.30 V, a few mA each). The 1v4 needs I2C at 100 kHz with this cable (now the default).
- Unplug any signal source from the LEMO inputs before connecting ASICs.
- In scripted menu sequences: RX UARTs all off first (`u`), then power, a 2 s wait (a monitor read right after power-on shows 0 mV), full reset, wait, configuration, wait, and channel 0 on (the second `u`) only when ready to receive. With channel 0 listening through reset and configuration, it often fills with random words.
- After a full reset, wait (0.2 s is enough) before configuring: the reset returns at once, and a configuration sent during it is lost. A pollable reset state register is set aside for later.

## 3 Configuration root

The configuration root (`pacman_menu`, `asic_config_root` in `asic_menu.c`) brings a chip up as the root of a chain:

| Register | Value | Meaning |
|---|---|---|
| 125 | 0x1 | downstream PISO enables: PISO 0 |
| 124 | 0x2 | upstream PISO enables: PISO 1 |
| 239, 240 | 0xFF | PISO drive: every TX slice enabled |
| 241, 242 | 0x00 | PISO drive current (`i_tx_diff`, active low): full |

The drive settings were added on 2026-10-31 (pacman-software 6d2ad8e) after the 1v4 results below.

# 4 Results

| Date | Board | Result |
|---|---|---|
| 2026-10-30 | 1v5 | Hello on UART 0 at the power-on drive: the expected reply, exactly. POSI request and PISO reply also decoded bit-exact from oscilloscope captures at the tile connector. |
| 2026-10-30 | 1v4 | At the power-on drive, the same wrong word on all four lanes (`0xFE000000 70387FC0`): the reply framed 7 bits late, with bit errors. All TX slices (239, 240 = 0xFF): exact on UARTs 0 and 1. Plus full current (241, 242 = 0x00): exact on all four. |
| 2026-10-31 | 1v4 | With 124 = 0x2 and full drive: exact on UART 0, and the built-in logic analyzer shows the request on POSI 0 and the exact reply on PISO 0, 8.2 µs later. PISO 2 and 3 (enabled neither up- nor downstream) stream junk into the receivers at full drive. |
| 2026-10-31 | 1v4 | Latest build (firmware 928fcf0, software 3c9b2d0), configuration root as built: the reply exact, twice. |
| 2026-11-01 | 1v4 | Software 671a3dd, the same sequence as a piped script: no reply (UART 0 streams junk). Cause: the configuration root sent right after the full reset is lost; the full reset returns at once, and the new menu input no longer sleeps between commands. With a 0.2 s pause after the full reset: the reply exact, twice. |
| 2026-11-01 | 1v4 | Software 290979d, a 16-BD DMA ring set up from the menu, with the 0.2 s pause: the reply exact, twice. |
| 2026-11-01 | 1v4 | Drive scan with the ASIC loopback test (800 readbacks of random register values per point): at the power-on current (241 = 0xCC) no slice setting (239 = 0x11 to 0xFF) passes; with all slices on, the current passes from 241 = 0x88 (0 errors) to 0x00, 0x99 is marginal (22 errors), 0xAA and weaker fail; one timeout at 0x22. Repeated with the RX buffer's heartbeat and rollover words off: the same threshold, and 0x22 passes (the timeout was those words in the stream). |
| 2026-11-01 | 1v5 | Old image (f516bc7), scripted: no reply and a stream of random words on UART 0, from the script's order (channel 0 on through reset and configuration; the monitor read before the regulators were up). By hand, and scripted in the order above: the reply exact (4 of 4); the register read-all valid throughout. |
| 2026-11-01 | 1v5 | Software 290979d: ASIC loopback 800 of 800 at full drive. Drive scan: every point passes except zero current (241 = 0xFF); one slice (239 = 0x11) passes at 241 = 0xCC, and the smallest current (241 = 0xEE) passes with all slices. The 1v5 needs far less drive than the 1v4. |
| 2026-11-02 | 1v5 | Configured by `larpix-lib`: the reply exact (results in larpix-lib `agents/larpix-lib.tex`). |

# 5 Open

- Systematic adjustment of the PISO drive (registers 239–242, and `v_cm_lvds_tx` 254–255) and of the PISO enables (124, 125): why the 1v4 needs far more drive than the 1v5 (which passes at one slice or the smallest current), why UARTs 2 and 3 need more than 0 and 1, and whether disabled PISOs at full drive are undriven or biased without data.
- The real bit layouts and defaults: a read-all of the registers on the 1v5, recorded (run by hand on 2026-11-01, not kept).
- ASIC stress tests on the 1v5 (later).
- LArPix v2b on the hw1v4: v2 clocks its UART at MCLK/2 (v3 at MCLK).